



Huayi Dynamics Technology (Dongguan) Co., Ltd

## Product specification

|                   |                 |
|-------------------|-----------------|
| Document Number   | Spec.GB4212-18V |
| Model             | GB4212-18V      |
| Name              | Robotic gearbox |
| Version           | A.0             |
| Prepared          | Li Jiwei        |
| Approved          | Xiao Hongwei    |
| Date              | 2024/12/03      |
| Customer Approved |                 |

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| Revision history |                 |         |            |          |              |
|------------------|-----------------|---------|------------|----------|--------------|
| No.              | Revision        | Version | Date       | Prepared | Approved     |
| 1                | Initial release | A.0     | 2024/12/03 | Li Jiwei | Xiao Hongwei |
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# 1 Application

This specification sheet is applicable to the GB4212-18V walking motor gearbox module, and the copyright belongs to Huayi Dynamics Technology (Dongguan) Co., Ltd.

# 2 Product specifications

| Seria<br>l<br>Num<br>ber | project                       | specifications                 | notes                                      |
|--------------------------|-------------------------------|--------------------------------|--------------------------------------------|
| 1                        | rated voltage                 | 18VDC                          |                                            |
| 2                        | no-load speed                 | 53±8% RPM                      |                                            |
| 3                        | no-load current               | 0.45A Max.                     |                                            |
| 4                        | rated load                    | 2.5N.m                         |                                            |
| 5                        | rated speed                   | 46±8% RPM                      |                                            |
| 6                        | Rated current                 | 1.3A Max.                      |                                            |
| 7                        | Insulation Class              | Class H                        |                                            |
| 8                        | Coil withstand<br>voltage     | DC500V、1sec                    | Leakage current less than 5mA              |
| 9                        | coil insulation               | DC500V                         | 50 MΩ Min                                  |
| 10                       | Usage temperature             | -20~80 °C                      |                                            |
| 11                       | Line resistance               | 3.24±8%ohm                     | 25°C                                       |
| 12                       | Line inductance               | 0.87±8% mH                     | 25°C                                       |
| 13                       | noise                         | ≤60 dB（A）                      | Test distance 100cm, back noise ≤ 32dB (A) |
| 14                       | Number of motor<br>pole slots | 8-pole 12 slot                 |                                            |
| 15                       | Winding<br>connection method  | Three phase star<br>connection |                                            |
| 16                       | reduction ratio               | 1/100.6                        |                                            |
| 17                       | life requirements             | >2200h                         |                                            |

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### 3 Reliability testing standards

| No. | Test project           | Describe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Judgment criteria                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.1 | lock rotor test        | <p>Working mode: Simulate motor operation mode.</p> <p>1 When the gearbox drives the tire to rotate, an impact stop test is conducted on the tire belt on both sides of the tire, and the stop stop device approaches at a speed of about TBDm/s,</p> <p>Locking current TBD A (can be achieved by using a cylinder close to the blocking stop).</p> <p>2 The cylinder stops for 2 seconds and then retracts after approaching the blocking and dragging disc, and repeats the blocking and stopping action every 3 seconds.</p> <p>3 Conduct a minimum of 1000 blockage tests</p>                                                                                                                                                                                                                                                                                                                                                                                                | The dental box as a whole has no abnormalities such as broken teeth, noise, abnormal sound, etc                                                                                                                                                                                                                                                                                                               |
| 3.2 | Temperature rise test  | <p>Working mode: Simulate motor operation mode. 1. Testing equipment: dynamometer loading;</p> <p>2 The whole machine is fixed on the testing fixture table, driven by an 18V voltage motor, with a load torque of 2.5N. m, 4N. m, 6N. m, and continuously running for 2 hours;</p> <p>3 Monitor the temperature of the motor enamel wire and the tooth box shell after reaching steady state.</p> <p>4 Environmental temperature 0/45/60 ° C</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | The maximum temperature under a load of 2.5N. m does not exceed 150 °C, and the product has no abnormal noise, etc                                                                                                                                                                                                                                                                                            |
| 3.3 | Temperature shock test | <p>Working mode: Power off mode. Relative humidity 93%</p> <p>1 Before the experiment, inspect the gearbox (appearance, noise, current, speed, forward and reverse rotation) according to the individual inspection case at room temperature to ensure that the individual meets the requirements, and record the test data.</p> <p>2 Place the test sample in a test chamber with a temperature equal to the laboratory temperature, at low temperature</p> <p>-Maintain for 3 hours at 20 °C± 2 °C, and temporarily maintain for 3 hours at 80 °C± 2 °C, with 5 cycles.</p> <p>3 The airflow velocity in the test chamber is 2.6m/s, and the temperature inside the test chamber is balanced.</p> <p>4 After the experiment is completed, the sample is allowed to recover at room temperature in the test chamber for 4 hours.</p> <p>5 After the experiment, inspect the motor (appearance Noise, current, speed, forward and reverse rotation, and record the test data;</p> | <p>1. The appearance surface shall not have defects such as cracks, wrinkles, brittleness, color transformation, etc., and the functional status shall meet Class A requirements (The functionality before and after testing is consistent with the design requirements);</p> <p>2 Noise, with a change of less than 5% before and after performance testing;</p> <p>3 The dental box has no oil leakage.</p> |
| 3.4 | Drop Test              | <p>Test the no-load speed and noise values of the gearbox on the entire machine before the drop test;</p> <p>The whole machine was dropped three times from a distance of 30cm from the cement ground;</p> <p>The whole machine fell 3 times from a horizontal distance of 80cm from the cement ground;</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>The appearance surface shall not show obvious damage, and the whole machine can work normally after testing;</p> <p>2. Noise and speed changes before and after testing are less than 5%.</p>                                                                                                                                                                                                              |

|     |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.5 | High temp& humidity running test | <p>Background environment: -20-80 °C @ 85% humidity, airflow velocity, circulation speed</p> <p>Operating at a rated load of 2.5N. m in a 2.6m/s environment; Testing method:</p> <p>1 Before the experiment, the appearance and load performance of each dental box should be tested at room temperature to ensure that the appearance, function, and performance indicators of the dental box meet the product requirements, and the test data should be recorded.</p> <p>2 Place the test sample in a -20 °C incubator, run at rated load for 24 hours, then shut down the incubator and the product. After the incubator temperature returns to room temperature, remove the product and let it stand at natural room temperature for 24 hours. Observe the appearance and test the performance data;</p> <p>3 After completing the b condition test, the product is placed in a 60 °C -85% humidity incubator. After running at rated load for 336 hours, the incubator and the product are turned off. After the incubator temperature returns to room temperature, the product is removed and left to stand at natural room temperature for 24 hours. The appearance is observed and performance data is recorded for testing;</p> <p>4 After completing the C condition test, the product is placed in an 80 °C -85% humidity incubator. After running at rated load for 80 hours, the incubator and the product are turned off. After the incubator temperature returns to room temperature, the product is removed and left at natural room temperature</p> <p>Let it stand for 24 hours, observe the appearance and test the performance, and record the data;</p> | <p>1 The appearance surface shall not have unacceptable defects such as wrinkling, warping deformation, delamination, embrittlement, cracking, surface contamination, precipitation, etc;</p> <p>2 After the completion of the b/c/d phase testing, the load performance, noise, and variation of the product shall not exceed 5%.</p> <p>3 There is no oil leakage in the dental box;</p> |
| 3.6 | Life test                        | <p>Background environment: 25 ± 5 °C,</p> <p>1 Before the experiment, the appearance, function, and performance indicators specified in the product development technical conditions of each dental box shall be tested at room temperature to ensure that the appearance, function, and performance indicators of the dental box meet the product requirements, and the test data shall be recorded.</p> <p>2 The gearbox is fixed on the testing fixture, and the output shaft is connected to the brake through an elastic coupling. The motor is controlled to operate through an external connector, and the brake is controlled by an external controller to adjust the load size.</p> <p>3 Voltage 18V, load speed closed-loop 47RPM ± 8%, load torque: 2.3N. m, load current TBDA</p> <p>4 The motor runs continuously for 2200 hours (1100 cycles) for 2 hours, with the addition of a fan for heat dissipation;</p> <p>5 After the experiment, inspect the gearbox (appearance, noise) with test cases at room temperature</p> <p>Sound, current, and rotational speed in both forward and reverse directions, and record the test data;</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>After completing the 1-life test, there was no failure, noise or abnormal sound in the posterior dental box Noise ≤ 65dB; 2. No oil leakage in the gearbox</p>                                                                                                                                                                                                                          |



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5. Performance curve (for reference only)

